

BME 140 W24

Lab #5 Report

Lab: B1-2

ECG Instrument

Brian Ngo

Due: 3/6/24

Introduction

This laboratory experiment focuses on the construction and characterization of an Electrocardiogram (ECG) amplifier system, aiming to amplify and accurately capture electrical signals produced by the heart. The primary goal is to design an amplifier with specific performance parameters, including gain and frequency response, to ensure reliable monitoring of cardiac activity. The major concepts addressed in this lab include the design and implementation of instrumentation amplifiers, bandpass filters, and non-inverting amplifiers, all crucial components for processing ECG signals.

Theoretical Background and Design

The construction of the ECG device entails achieving a voltage gain of 1000 and a frequency response spanning from 0.01 to 100 Hz. The procedure involves two main stages, starting with an instrumentation amplifier targeting a gain of 40. To achieve this gain, the initial differential gain of 5 mandates a resistor value of **10 kΩ for R1**. Subsequently, setting **R2 at 20 kΩ** yields a second differential gain of 8. Utilizing the gain equation,

$$G = (1 + 2R2/R1) * R4/R3$$

with **R3 and R4 values of 10 kΩ and 80 kΩ respectively**, assures a buffered input impedance approaching infinity.

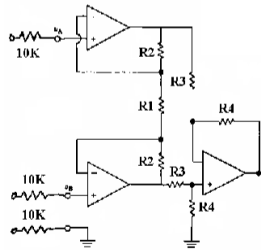


Figure 1: Instrumentation Amplifier

The subsequent stage incorporates a band-pass filter with a gain of 25, facilitated by a non-inverting amplifier. Determining resistor values using the following,

$$G = 1 + R6/R7$$

a **0.01 μF capacitor (C5)** and a **1 μF capacitor (C6)** establish cutoff frequencies. By employing the following frequency cutoff equation,

$$F_c = 1/(2\pi RC)$$

R5 is calculated at 15.9 MΩ and R6 at 159 kΩ. A total resistance of 15.9 MΩ comprises the sum of three resistors: 10 MΩ, 5.6 MΩ, and 330 kΩ, for R5. With R7 set at 6.63 kΩ, an additional R5 is integrated to mitigate potential bias current. Subsequently, a voltage divider, featuring 100 kΩ and 1 kΩ resistors, facilitates attenuation of the signal by a factor of 1/100,

preventing significant signal loss. This step ensures effective signal transmission through subsequent stages while preserving signal integrity.

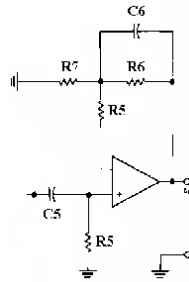


Figure 2: Band Pass Filter

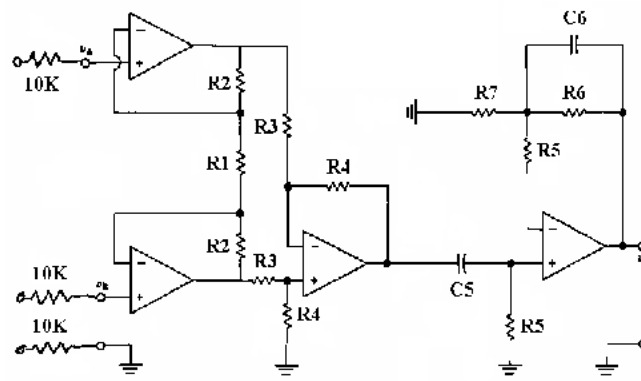


Figure 3: ECG Amplifier

Results and Discussion

- What is the common mode gain at 10 Hz?

The common mode gain was 0.001, which is expected because it should be near 0.

- Plot the gain curve.

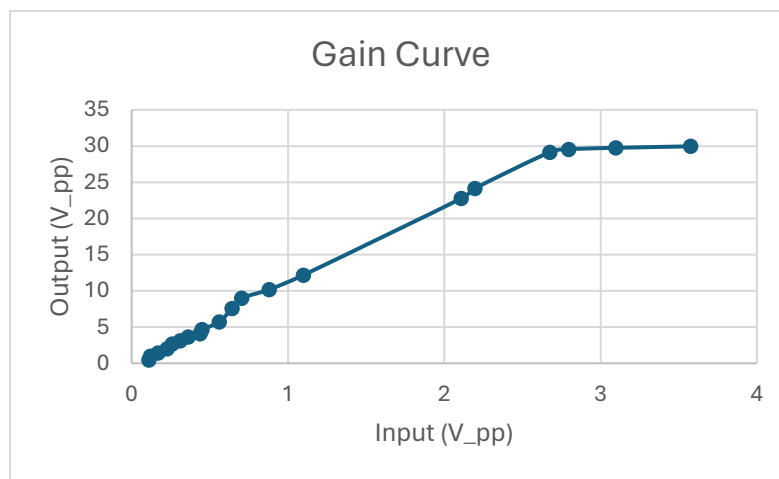


Figure 4: Gain Curve at 10 Hz

To calculate differential gain, it is simply the gain from the instrumentation amplifier times the gain from the bandpass filter.

$$\text{Instrumentation Amplifier Gain} = \frac{17.17}{0.430} = 39.93$$

$$\text{Band Pass Filter} = \frac{11.58}{0.4481} = 25.84$$

So, our differential gain is 1031.

- *Compare it to theoretical prediction. What is the percentage difference?*

The theoretical gain is 1000 ($40 * 25$). Our measured differential gain is relatively like the predicted gain, and the error is $(|1031 - 1000| / 1000) * 100 = 3.1\%$

- *What is the cause of the prediction error?*

The primary cause of prediction error is because of the imperfect tolerances and values for the resistors, capacitors, and op-amps.

- *At what input voltage does the output start to saturate? What is the working differential input range of this ECG device?*

At around 14.5 mV, we started to experience saturation at 15 V. With the minimum input of 1 mV, our dynamic range varied up to 100 mV.

- *What is the common mode rejection ratio? Compare it to the prediction in your homework assuming there is one resistor pair mismatch of 10%.*

The common mode gain is 0.001 ± 0.001 V (the value should be close to 0). To find CMRR, we divide our differential gain (1031) by 0.01 to yield $1.031 * 10^6$. After taking the log ($= 20\log$), we get 120.27 dB. This was much greater than the predicted value of 98.5 with a 10% resistor mismatch for the first stage, showing the efficacy of the band-pass filter in negating the effects of resistor mismatch while increasing differential gain. We checked all resistors and capacitors with DMM, though most were within a certain tolerance range of the exact value, and thus we expected error and CMRR to be approximately what we got.

- *Plot the measured amplitude as a function of frequency from the bode plot. When signal amplitude has fallen by -3 dB, what is the low pass cutoff frequency?*

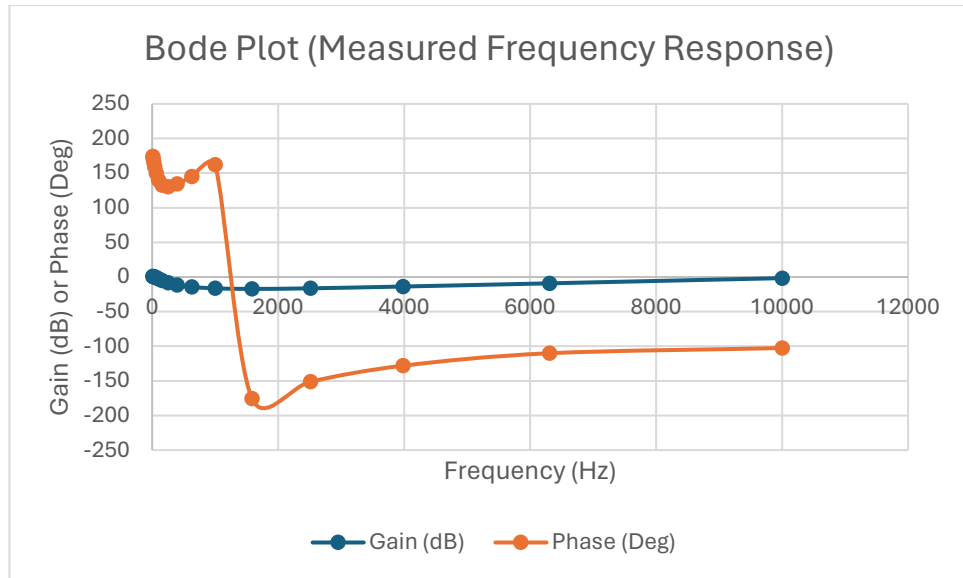


Figure 5: Measured Frequency Response Bode Plot

When the amplitude has fallen by -3 dB, the low cut-off frequency is approximately 100 Hz.

- Plot the calculated frequency response for the filter. Does the measured curve look like the calculated curve (at least in the high-frequency range)? Show the procedure of your calculation.

$$T(f) = \frac{G}{G_o} \text{Frequency Response}$$

$$T(f) = \left(1 + \frac{R_6}{R_7} \frac{1}{\left(1 + \frac{jf}{f_{lc}} \right)} \right) \left(\frac{\frac{jf}{f_{hc}}}{1 + \frac{jf}{f_{lc}}} \right) = 20 \log(T(f)) = 20 \log \left(\frac{G}{1000} \right)$$

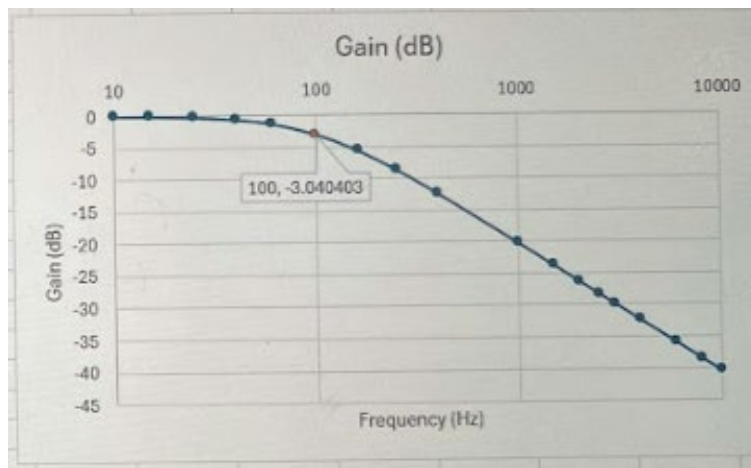


Figure 6: Calculated Frequency Response Bode Plot (from lab partner)

The measured curve looks like the high-frequency calculated curve. The calculation required a transfer function and a logarithm.

- *Generate a table for the specifications of your ECG device.*

Input Signal Dynamic Range (mV)	DC Offset Voltage (mV)	Frequency Response Range (Hz)	Differential Gain	CMRR (dB)
1 mV – 100 mV	-60 ± 30 mV	0.01 - 150 Hz	1031	120.27

Conclusion

This laboratory experiment provided valuable insights into the construction and performance evaluation of an Electrocardiogram (ECG) amplifier system. Through the careful assembly of amplifier and filter components, coupled with the implementation of voltage dividers to manage gain and cutoff frequencies, we successfully created a system capable of detecting heart rhythms. By addressing impedance considerations and employing high resistance values, we enhanced circuit stability and minimized signal distortion. While our results deviated from ideal expectations, with a differential gain of approximately 40 in the first step and 25 in the second step, we observed a notable improvement in the common-mode rejection ratio (CMRR). This enhancement underscored the filter's ability to bolster differential gain while mitigating common-mode interference, critical for accurate ECG signal detection. Despite the challenges encountered, this experiment provided invaluable firsthand experience in biomedical electronics, fostering a deeper understanding of signal-processing techniques essential for healthcare applications.